On covering radius of generalized Zetterberg codes*

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Abstract

In this conference paper we study the covering radius of the generalized Zetterberg codes $C_s(q_0)$ of length q_0^s+1 and dimension q_0^s+1-2s over the finite field $\mathbb{F}_{q_0}$ where q_0 is even. We prove that $\rho(C_s(q_0)) \leq 3$, and determine its exact value for a wide range of s for each such q_0.

Key Words : Covering radius. Zetterberg code, Weil bound, character sums.

1 Introduction

1.1 The covering radius

Let $\mathbb{F}_{q_0}$ denote the finite field of q_0 elements, where q_0 is a prime power. Let n be a positive integer. For two vectors $\boldsymbol{x}=(x_1,\cdots,x_n), \boldsymbol{y}=(y_1,\cdots,y_n) \in \mathbb{F}_{q_0}^n$, their *Hamming distance* is

$$d_{\mathrm{H}}(\boldsymbol{x},\boldsymbol{y}) = \#\{\, i : x_i \neq y_i \,\}.$$

Here $\#I$ denotes the cardinality of a finite set I. Any subset $C \subseteq \mathbb{F}_{q_0}^n$ is called a *code* of length n over $\mathbb{F}_{q_0}$. The *covering radius* of C is defined as

$$\rho(C) = \max_{\boldsymbol{x}\in\mathbb{F}_{q_0}^n} d_{\mathrm{H}}(\boldsymbol{x},C), \quad d_{\mathrm{H}}(\boldsymbol{x},C) = \min_{\boldsymbol{c}\in C} d_{\mathrm{H}}(\boldsymbol{x},\boldsymbol{c}).$$

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The covering radius is a fundamental concept in coding theory and discrete mathematics. It arises naturally in **decoding**, where $\rho(C)$ represents the maximum number of errors a received word may contain before becoming ambiguous under nearest-neighbor decoding; in **data compression** and **quantization**, where it corresponds to the worst-case distortion under nearest-codeword approximation and relates to sphere-covering problems; and in **testing**, **fault detection**, and **memory coding**, where it measures the completeness of test sets and the robustness of storage schemes. Comprehensive treatments of these topics can be found in the standard monographs and surveys [BLP98, CHLL97, CKMS85, CLLM97].

It is well known that $\left\lfloor \frac{d(C)-1}{2} \right\rfloor \leq \rho(C)$. A code satisfying $\rho(C) = \left\lfloor \frac{d(C)-1}{2} \right\rfloor$ is called *perfect*; such codes partition the Hamming space into disjoint Hamming spheres and simultaneously achieve optimal error-correction capability and maximum packing efficiency. When $\rho(C) = \left\lfloor \frac{d(C)-1}{2} \right\rfloor + 1$, the code is termed *quasi-perfect*. The parameters of all perfect codes over finite fields were completely determined in the 1970s [Tie73, Tie74], and the classification of linear perfect codes is now fully known (see [HP03]). By constrast, the classification of quasi-perfect codes remains substantially more complicated. Consequently, recent research has focused on constructing explicit families of quasi-perfect codes with diverse parameters (see, e.g., [DD08, DDR11, Dod86, Dod85, EM05, GPZ60, Hel78, LH16]); see also [SHÖS22, SHÖ23, SLHÖ25] for several of the latest developments.

Despite extensive study, determining the covering radius of a general linear code remains a challenging problem—it is both **NP-hard** and **co-NP-hard** [McL84]. Consequently, most work has focused on deriving upper and lower bounds rather than exact values [AB02, Hel85, MC03, Slo86, BS17, BS18, Fra05, FC04, Coh97, VS89]). Exact covering radii are known only for a few highly structured families of codes; see, for example, [DD08, DDR11, Dod86, Dod85, GPZ60, Hel78, Hou96, LH16]. For most recent progress, we refer the reader to [GKLW23, GL24, DMT22, SLHÖ25].

1.2 Summary of Main Results

Recent research [SHÖ23, SLHÖ25] has provided a detailed account of the covering radius of generalized Zetterberg codes $C_s(q_0)$ for odd q_0. Although these works established impressive results, their proofs rely on lengthy and intricate elementary arguments, and the *even-field case* (q_0 even) has remained largely open.

In this paper we employ analytic number-theoretic techniques—specifically, character sums and Weil-type estimates—to study the covering radius of $C_s(q_0)$ for even q_0. The main result can be summarized as follows:

Theorem 1. *Let q_0 be even. The covering radius of $C_s(q_0)$ is at most three and is given by*

i). If $s = 1$, then $\rho(C_s(q_0)) = 1$;

ii). If $s = 2$, then $\rho(C_s(q_0)) = 2$;

iii). If $s \geq 4$ is even, then $\rho(C_s(q_0)) = 3$;

iv). If $s \geq 3$ is odd, and $s \leq s^*(q_0)$, then $\rho(C_s(q_0)) = 2$;

v). If $s \geq 3$ is odd, and $s \geq s_*(q_0)$, then $\rho(C_s(q_0)) = 3$;

vi). If $s \geq 3$ is odd, and $s'|s$ with $\rho(C_{s'}(q_0)) = 3$, then $\rho(C_s(q_0)) = 3$.

Here $s_*(q_0) \geq 3$ (if exists) is the least odd number s satisfying the inequality

$$q_0^s - q_0^{s/2}\left(\frac{q_0}{2}\right)^{q_0-1}\left(2q_0^2 - 7q_0 + 3\right) > \left(\frac{q_0}{2}\right)^{q_0-1}\left(4q_0 - 2\right) - q_0^2,$$

and $s^*(q_0) \geq 3$ (if exists) is the largest odd number satisfying $s^*(q_0) \leq q_0/2$.

It is easy to check that $s_*(2) = 3, s_*(4) = 7$, and if $q_0 \geq 8$, then $s_*(q_0)$ is an odd number in the range

$$2q_0 + 2 - \frac{2(q_0-1)\log 2}{\log q_0} < s_*(q_0) < 2q_0 + 4 - \frac{2(q_0-2)\log 2}{\log q_0}.$$

In Table 1 below, we list the values $s^*(q_0)$ and $s_*(q_0)$ for $q_0 = 2^t$ where $1 \leq t \leq 7$. We also list the missing odd numbers s in the interval $(s^*(q_0), s_*(q_0))$ where the value $\rho(C_s(q_0))$ is not yet determined.

Table 1: $s^*(q_0), s_*(q_0)$ and values not covered by Theorem 1.

$q_0 = 2^t$	s^*	s_*	the set of missing odd numbers in (s^*, s_*)
2		3	$\emptyset$
4		7	{3,5}
8	3	15	{5,7,9,11,13}
16	7	27	odd numbers from 9 to 25
32	15	55	odd numbers from 17 to 53
64	31	111	odd numbers from 33 to 109
128	63	223	odd numbers from 65 to 221

The computations performed using Magma agree perfectly with Proposition 1. For the missing odd s in $(s^*(q_0), s_*(q_0))$, for example, for $q_0 = 4$, we have $\rho(C_3(4)) = 2$ and $\rho(C_5(4)) = 3$; and for $q_0 = 8$, $\rho(C_5(8)) = \rho(C_7(8)) = 2$.

This paper is organized as follows. Section 2 introduces the notation. In Section 3 we define the generalized Zetterberg code $C_s(q_0)$ in even characteristic. In Section 4 we provide a detailed proof that $\rho(C_s(q_0)) \leq 3$ for even q_0 and all s, using character sums and Weil-type estimates. In Section 5 we provide a criterion to determine the exactly value $\rho(C_s(q_0))$ for even q_0 from which Proposition 1 can be derived.

2 Notation

We use the following notation throughout the paper.

- q_0 denotes a prime power, which we assume to be even, s a positive integer, $q = q_0^s$;
- $\mathbb{F}_{q_0}, \mathbb{F}_q$ and $\mathbb{F}_{q^2}$ are the finite fields of order q_0, q and q^2 respectively;
- $\mathbb{F}_q^* := \mathbb{F}_q \setminus \{0\}$;
- $H \subseteq \mathbb{F}_{q^2}$ is the multiplicative subgroup of order $q+1$, defined by

$$H := \left\{x \in \mathbb{F}_{q^2} : x^{q+1} = 1\right\};$$

- For any $x \in \mathbb{F}_{q^2}$, write $\overline{x} := x^q$;
- $C_s(q_0)$ denotes the q_0-ary generalized Zetterberg code defined in (1);
- $\mathbb{P}^1(\mathbb{F}_q) = \mathbb{F}_q \cup \{\infty\}$.

3 Generalized Zetterberg in even characteristic

Let ξ be a generator of H so that

$$H = \left\{1, \xi, \xi^2, \cdots, \xi^q\right\}.$$

The q_0-ary *generalized Zetterberg code*, denoted by $C_s(q_0)$, is defined as

$$C_s(q_0) = \left\{(c_0, c_1, \cdots, c_q) \in \mathbb{F}_{q_0}^{q+1} : \sum_{i=0}^{q} c_i \xi^i = 0\right\}. \tag{1}$$

Thus $C_s(q_0)$ is a q_0-ary linear code of length $q+1$.

Let $h_{q_0}(X) \in \mathbb{F}_{q_0}[X]$ be the minimal polynomial of ξ over $\mathbb{F}_{q_0}$. Since $\mathbb{F}_{q^2} = \mathbb{F}_q(\xi)$, we have

$$[\mathbb{F}_q(\xi) : \mathbb{F}_{q_0}] = [\mathbb{F}_q(\xi) : \mathbb{F}_q] \cdot [\mathbb{F}_q : \mathbb{F}_{q_0}] = 2s,$$

so $\deg h_{q_0}(X) = 2s$ and $h_{q_0}(X) | \left(x^{q+1} - 1\right)$.

Each $\underline{c} = (c_0, \ldots, c_q) \in \mathbb{F}_{q_0}^{q+1}$ corresponds to a polynomial $\underline{c}(X) = c_0 + c_1 X + \cdots + c_q X^q$. This gives an isomorphism from the vector space $\mathbb{F}_{q_0}^{q+1}$ to the quotient ring $\mathbb{F}_{q_0}[X] / \left(X^{q+1} - 1\right)$. It is easy to see that

$$\underline{c} \in C_s(q_0) \Longleftrightarrow \underline{c}(\xi) = 0 \Longleftrightarrow h_{q_0}(X) | \underline{c}(X),$$

hence $C_s(q_0)$ can be identified with the cyclic group in $\mathbb{F}_{q_0}[X]/\left(X^{q+1}-1\right)$ generated by $h_{q_0}(X)$. Since $\deg h_{q_0}(X)=2s$, we have

$$\dim C_s(q_0)=q+1-2s,$$

so the code $C_s(q_0)$ has parameters $[q+1,q+1-2s]_{q_0}$.

By the code parameters of $C_s(q_0)$ and the sphere-packing bound (see [HP03, MS77])

$$A_{q_0}(n,d)\leq \frac{q_0^n}{\sum_{i=0}^{\lfloor \frac{d-1}{2}\rfloor}\binom{n}{i}(q_0-1)^i},$$

where $A_{q_0}(n,d)$ denotes the largest cardinality of a q_0-ary code of length n with minimum distance d, we can easily obtain the following:

$$d\left(C_s(q_0)\right)\leq 4,\quad \forall\, q_0\geq 4 \text{ even.} \tag{2}$$

It is known that $\rho\left(C_s(q_0)\right)$ can be described in a very simple way (see [CHLL97, Hel85] and [SHÖ23]):

Lemma 1. *The covering radius of $C_s(q_0)$ is the least positive integer ρ such that for any $\alpha\in\mathbb{F}_{q^2}$, there exist $(c_1,\ldots,c_\rho)\in\mathbb{F}_{q_0}^\rho$ and $(x_1,\ldots,x_\rho)\in H^\rho$ such that*

$$\alpha=c_1x_1+\cdots+c_\rho x_\rho.$$

4 Covering radius of $C_s(q_0)$ is at most 3

Denote by $\rho\left(C_s(q_0)\right)$ the covering radius of the code $C_s(q_0)$. We first prove a technical lemma, which works for both even and odd q_0.

Lemma 2. *For any $\alpha\in\mathbb{F}_{q^2}$ and any $(c_1,c_2,c_3)\in\mathbb{F}_{q_0}^3$, denote by N_1 the number of solutions $(x_1,x_2,x_3)\in H^3$ to the equation*

$$\alpha=c_1x_1+c_2x_2+c_3x_3. \tag{3}$$

Denote by N_2 the number of solutions $(x_1,x_2)\in H^2$ to the equation

$$c_2(\overline{\alpha}-c_1\overline{x}_1)x_2^2-\left((\alpha-c_1x_1)(\overline{\alpha}-c_1\overline{x}_1)+c_2^2-c_3^2\right)x_2+c_2(\alpha-c_1x_1)=0. \tag{4}$$

If $c_3\neq 0$, then $N_1=N_2$.

Proof. Suppose $(x_1,x_2,x_3)\in H^3$ is a solution to Eq (3), then we have

$$c_3x_3=\alpha-c_1x_1-c_2x_2.$$

Since $c_i\in\mathbb{F}_{q_0}\subseteq\mathbb{F}_q$ for each i, we have

$$c_3\overline{x}_3=\overline{\alpha}-c_1\overline{x}_1-c_2\overline{x}_2.$$

Here for $x \in \mathbb{F}_{q^2}$ we denote $\overline{x} := x^q$. Since $x_3 \in H$, $\overline{x}_3 x_3 = 1$, so we have

$$c_3^2 = (\alpha - c_1x_1 - c_2x_2)(\overline{\alpha} - c_1\overline{x}_1 - c_2\overline{x}_2). \tag{5}$$

From this we obtain

$$(\alpha - c_1x_1)(\overline{\alpha} - c_1\overline{x}_1) - c_2x_2(\overline{\alpha} - c_1\overline{x}_1) - c_2\overline{x}_2(\alpha - c_1x_1) + c_2^2 = c_3^2. \tag{6}$$

Multiplying x_2 on both sides of Eq (6) and noting $x_2\overline{x}_2 = 1$, we obtain Eq (4).

On the other hand, suppose $(x_1, x_2) \in H^2$ is a solution to Eq (4), then (x_1, x_2) also satisfies Eq (6) and hence Eq (5), so $\alpha - c_1x_1 - c_2x_2 \neq 0$ as $c_3 \neq 0$. It is easy to see that by defining

$$x_3 := \frac{\alpha - c_1x_1 - c_2x_2}{c_3},$$

we have $x_3 \in H$ and this triple $(x_1, x_2, x_3) \in H^3$ is a solution to Eq (3). This completes the proof of Lemma 2. $\square$

Theorem 2. *Let $q_0 \geq 2$ be even and $s \geq 1$. Then the covering radius of $C_s(q_0)$ is at most* 3.

Proof. The case $s = 1$ is trivial, because in this case $q = q_0$, by the standard polar representation, every $x \in \mathbb{F}_{q_0^2}^*$ can be written uniquely as $x = yz$ with $y \in \mathbb{F}_{q_0}^*$ and $z \in H$. Therefore in this case $\rho(C_1(q_0)) = 1$.

If $q_0 = 2$ and $s = 2$, it is easy to check by MAGMA that $\rho(C_s(2)) = 2$. So from now on let us assume either "$q_0 = 2$ and $s \geq 3$" or "$q_0 \geq 4$ and $s \geq 2$", so we always have $q \geq 8$.

By Lemma 1, to show that $\rho(C_s(q_0)) \leq 3$, it suffices to prove that for every $\alpha \in \mathbb{F}_{q^2}$, there exist $c_1, c_2, c_3 \in \mathbb{F}_{q_0}$ and $x_1, x_2, x_3 \in H$ such that

$$\alpha = c_1x_1 + c_2x_2 + c_3x_3. \tag{7}$$

We may write α as $\alpha = \alpha_0\alpha_H$ with $\alpha_0 \in \mathbb{F}_q$ and $\alpha_H \in H$. Suppose (7) holds for α. Then

$$\alpha_0 = c_1\frac{x_1}{\alpha_H} + c_2\frac{x_2}{\alpha_H} + c_3\frac{x_3}{\alpha_H},$$

where $\frac{x_i}{\alpha_H} \in H$ for each i since H is the multiplicative subgroup of order $q+1$ in $\mathbb{F}_{q^2}^*$. Therefore, without loss of generality, we may assume $\alpha \in \mathbb{F}_q$. If $\alpha \in \mathbb{F}_{q_0}$, we can set $c_1 = \alpha$, $x_1 = 1$, and $c_2 = c_3 = 0$, so (7) holds trivially. Hence, from now on let us assume that $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$.

For fixed $(c_1, c_2, c_3) \in \mathbb{F}_{q_0}^{*3}$, let us define

$$N(c_1, c_2, c_3) = \#\left\{(x_1, x_2, x_3) \in H^3 : \alpha = c_1x_1 + c_2x_2 + c_3x_3\right\}.$$

Our goal is to show that $N(c_1, c_2, c_3) > 0$ for any $(c_1, c_2, c_3) \in (\mathbb{F}_{q_0}^*)^3$ and any $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$. This would imply that $\rho\left(C_s(q_0)\right) \leq 3$.

By Lemma 2 again, $N(c_1, c_2, c_3)$ equals the number of solutions $(x_1, x_2) \in H^2$ to the equation

$$c_2\left(\alpha + c_1\overline{x}_1\right)x_2^2 + \left((\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2\right)x_2 + c_2\left(\alpha + c_1x_1\right) = 0. \tag{8}$$

For any fixed $x_1 \in H$, to analyze the existence of solutions $x_2 \in H$ of Eq (8), we denote

$$\triangle(x_1) := (\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2.$$

We consider two cases.

Case 1: $\triangle(x_1) = 0$**.** Then (8) reduces to

$$(\alpha + c_1\overline{x}_1)c_2x_2^2 + (\alpha + c_1x_1)c_2 = 0,$$

that is,

$$x_2^2 = \frac{\alpha + c_1x_1}{\alpha + c_1\overline{x}_1}.$$

Since q is even, this gives a unique solution $x_2 \in H$ to Eq (8).

Case 2: $\triangle(x_1) \neq 0$**.** It is known that for each fixed $x_1 \in H$, Eq (8) has one and hence two distinct solutions $x_2 \in H$ if and only if

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2^2(\alpha + c_1\overline{x}_1)(\alpha + c_1x_1)}{\left((\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2\right)^2}\right) = 1. \tag{9}$$

Note that the left hand side can be rewritten as

$$\begin{aligned}
&\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2^2\left((\alpha + c_1\overline{x}_1)(\alpha + c_1x_1) + c_2^2 + c_3^2\right) + c_2^2\left(c_2^2 + c_3^2\right)}{\left((\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2\right)^2}\right)\\
=&\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2^2}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right) + \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2^2\left(c_2^2 + c_3^2\right)}{\left((\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2\right)^2}\right)\\
=&\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2^2}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right) + \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2(c_2 + c_3)}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right)\\
=&\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2c_3}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right).
\end{aligned}$$

So the following condition is equivalent to (9):

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}\left(\frac{c_2c_3}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right) = 1.$$

Hence for the given $x_1 \in H$, the number of solutions $x_2 \in H$ to Eq (8) is given by

$$1 - \psi\left(\frac{c_2c_3}{(\alpha + c_1x_1)(\alpha + c_1\overline{x}_1) + c_2^2 + c_3^2}\right),$$

where $\psi : \beta \mapsto (-1)^{\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(\beta)}$ is the standard additive character defined on $\mathbb{F}_q$. Summarizing **Cases 1** and **2** above, we conclude that

$$\begin{aligned} N(c_1, c_2, c_3) &= \sum_{\substack{x_1 \in H \\ \triangle(x_1)=0}} 1 + \sum_{\substack{x_1 \in H \\ \triangle(x_1)\neq 0}} \left(1 - \psi\left(\frac{c_2 c_3}{(\alpha + c_1 x_1)(\alpha + c_1 \overline{x}_1) + c_2^2 + c_3^2}\right)\right) \\ &= q + 1 - \sum_{\substack{x_1 \in H \\ \triangle(x_1)\neq 0}} \psi\left(\frac{c_2 c_3}{(\alpha + c_1 x_1)(\alpha + c_1 \overline{x}_1) + c_2^2 + c_3^2}\right). \end{aligned}$$

Let us define

$$T := \sum_{\substack{x_1 \in H \\ \triangle(x_1)\neq 0}} \psi\left(\frac{c_2 c_3}{(\alpha + c_1 x_1)(\alpha + c_1 \overline{x}_1) + c_2^2 + c_3^2}\right).$$

We fix an element $a \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ and consider the bijection $\phi : \mathbb{P}^1(\mathbb{F}_q) \to H$ given by $\phi : y \mapsto \frac{y+a}{y+\overline{a}}$. Since $\phi(\infty) = 1 \in H$ and

$$\triangle(1) = (\alpha + c_1 + c_2 + c_3)^2 \neq 0,$$

using this bijection ϕ we can rewrite T as

$$T = \psi\left(\frac{c_2 c_3}{(\alpha + c_1 + c_2 + c_3)^2}\right) + \sum_{\substack{y \in \mathbb{F}_q \\ \triangle(\phi(y))\neq 0}} \psi\left(\frac{c_2 c_3}{(\alpha + c_1 \phi(y))\,(\alpha + c_1 \phi(y)^{-1}) + c_2^2 + c_3^2}\right).$$

The right hand side can be rewritten as

$$T = \psi\left(\frac{c_2 c_3}{(\alpha + c_1 + c_2 + c_3)^2}\right) + \sum_{y \in \mathbb{F}_q \setminus S} \psi\left(f(y)\right),$$

where $f(y)$ is a rational function in $\mathbb{F}_q(y)$ given by

$$f(y) = \frac{c_2 c_3}{(\alpha + c_1 + c_2 + c_3)^2}\left(1 + \frac{A}{y^2 + (a + \overline{a})\,y + a\overline{a} + A}\right),$$

S is the set of poles of $f(y)$ in $\mathbb{F}_q$ and

$$A = \frac{\alpha c_1\,(a + \overline{a})^2}{(\alpha + c_1 + c_2 + c_3)}.$$

Now we apply [CP06, Theorem 1.1]. For the benefit of readers, we include the full statement here:

Lemma 3. *[CP06] Consider the finite field $\mathbb{F}_q$ with characteristic p. Let $f(X) \in \mathbb{F}_q(X)$ be a rational function,*

$$f(X) = p(X) + \frac{r(X)}{q(X)}, \quad \deg(r(X)) < \deg(q(X)), \quad M := \deg(p(X)),$$

where $p(X), q(X), r(X) \in \mathbb{F}_q[x]$ are polynomials, and $\gcd(r(X), q(X)) = 1$. Suppose further that $f(X)$ is non-constant, and that $M = 0$ or $p \nmid M$. Write

$$q(X) = \prod_{i=1}^{Q} q_i(X)^{m_i},$$

where $q_i(X)$ are distinct irreducible polynomials over $\mathbb{F}_q$, and $m_i \geq 1$ for each i. Define

$$L := \sum_{i=1}^{Q} (m_i + 1) \deg(q(X)).$$

Then there exist complex numbers $\omega_j \in \mathbb{C}$, $1 \leq j \leq M + L - 1$, such that

$$\sum_{\beta \in \mathbb{F}_q \setminus S} \psi(f(\beta)) = - \sum_{j=1}^{M+L-1} \omega_j,$$

where S is the set of poles of $f(X)$. Additionally, $|\omega_j| = \sqrt{q}$ for all j, except for a single value, j', satisfying $|\omega_{j'}| = 1$. Thus,

$$\left| \sum_{\beta \in \mathbb{F}_q \setminus S} \psi(f(\beta)) \right| \leq 1 + (M + L - 2) \sqrt{q}. \qquad \square$$

From the explicit form of $f(y)$, it is easy to see that $M = 0$ and $L = 4$, so we deduce

$$|T| \leq 1 + (1 + 2\sqrt{q}) = 2\,(1 + \sqrt{q})\,.$$

Armed with this estimate, we obtain

$$N(c_1, c_2, c_3) \geq q + 1 - |T| \geq q + 1 - 2\,(1 + \sqrt{q}) = q - 2\sqrt{q} - 1.$$

Since we have assumed that $q \geq 8$, it is easy to check that $q - 2\sqrt{q} - 1 > 0$, so $N(c_1, c_2, c_3) > 0$ as desired for any $(c_1, c_2, c_3) \in \mathbb{F}_{q_0}^{*3}$ and any $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$. This completes the proof of Theorem 2.

5 Determining covering radius of $C_s(q_0)$: q_0 even

We know by the proof of Theorem 2 that $\rho(C_s(q_0)) = 2$ if and only if for any $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$ (note that the case $\alpha \in \mathbb{F}_{q_0}$ is trivial), there exist $(c_1, c_2) \in \mathbb{F}_{q_0}^{*2}$ and $(x_1, x_2) \in H^2$ such that

$$\alpha = c_1 x_1 + c_2 x_2. \tag{10}$$

Carefully analyzing this equation for solutions $x_1, x_2 \in H$, we can obtain the following criterion

Theorem 3. *Let $q_0 \geq 2$ be even and $s \geq 2$. Then $\rho(C_s(q_0)) = 3$ if and only if there exists $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$ such that*

1. $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}}(\alpha) = 0$, *and*
2. $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}}\left(\frac{1}{1+b\alpha}\right) \in \{0,1\}$ *for all $b \in \mathbb{F}_{q_0}^*$.*

Otherwise $\rho(C_s(q_0)) = 2$.

Now we use Theorem 3 to prove iii) of Proposition 1. We leave others to arXiv:2510.25295.

Corollary 1. *Let q_0 be even and $s \geq 2$ be a positive integer. If $s = 2$ or if $s \geq 3$ is odd and $s \leq \frac{q_0}{2}$, then $\rho(C_s(q_0)) = 2$.*

Proof. If $s = 2$ then $q = q_0^2$, for any $\alpha \in \mathbb{F}_q$,

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}}(\alpha) = \alpha + \alpha^{q_0} = 0 \quad \Longrightarrow \quad \alpha \in \mathbb{F}_{q_0}.$$

So there is no $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$ such that $\mathrm{Tr}_{q_0}^q(\alpha) = 0$, by Theorem 3, $\rho(C_s(q_0)) = 2$.

Now suppose s is odd and $s \leq q_0/2$. For any $\alpha \in \mathbb{F}_q \setminus \mathbb{F}_{q_0}$, let t be the least positive integer such that $\alpha^{q_0^t} = \alpha$. It is easy to see that $t|s$ and $\alpha, \ldots, \alpha^{q_0^{t-1}}$ are distinct elements in $\mathbb{F}_{q_0^t} \subseteq \mathbb{F}_q$. Define the rational function $f(x) \in \mathbb{F}_q(x)$ as

$$f(x) := \sum_{j=0}^{s-1} \frac{1}{1 + x\alpha^{q_0^j}}.$$

Let $s = t \cdot l$. Since each integer j such that $0 \leq j \leq s-1$ can be written uniquely as $j = t \cdot k + i$ where $0 \leq k \leq l-1$ and $0 \leq i \leq t-1$, we have

$$\begin{aligned} f(x) &= \sum_{i=0}^{t-1} \sum_{k=0}^{l-1} \frac{1}{1 + x\alpha^{q_0^{tk} q_0^i}} \\ &= \frac{s}{t} \left(\sum_{i=0}^{t-1} \frac{1}{1 + x\alpha^{q_0^i}} \right) = \sum_{i=0}^{t-1} \frac{1}{1 + x\alpha^{q_0^i}}. \end{aligned}$$

Let us define

$$g(x) = \sum_{i=0}^{t-1} \frac{1}{1 + x\alpha^{q_0^i}}.$$

Then $g(x)$ is a non-constant rational function defined over $\mathbb{F}_q$. We can write

$$g(x) = \frac{N(x)}{D(x)},$$

where $D(x), N(x) \in \mathbb{F}_q[x]$ are given by

$$D(x) = \prod_{i=0}^{t-1} \left(1 + x\alpha^{q_0^i}\right), \quad N(x) = D(x) \left(\sum_{i=0}^{t-1} \frac{1}{1 + x\alpha^{q_0^i}}\right).$$

It is easy to see that $\deg D = t$ and $\deg N \leq t-1$, and $N \neq 0$.

For any ϵ, denote by N_ϵ the number of solutions $x \in \mathbb{F}_{q_0}$ such that $f(x) = g(x) = \epsilon$, which is equivalent to the equation

$$N(x) + \epsilon D(x) = 0.$$

Since $N(x), D(x) \in \mathbb{F}_q[x]$ are not zero polynomials with $\deg D = t$, $\deg N \leq t-1$, we have

$$N_0 + N_1 \leq t - 1 + t = 2t - 1 \leq 2s - 1 \leq q_0 - 1,$$

which implies that there is always $b \in \mathbb{F}_{q_0}$ such that

$$\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_{q_0}} \left(\frac{1}{1+b\alpha}\right) = \frac{1}{1+b\alpha} + \frac{1}{1+b\alpha^{q_0}} + \cdots + \frac{1}{1+b\alpha^{q_0^{s-1}}} = f(b) \notin \{0,1\}.$$

Obvioulsy $b \neq 0$. Thus by Theorem 3, $\rho(C_s(q_0)) = 2$. $\square$

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